

UNIT-I

Introduction to Classical Reinforcement Learning

1. Understanding Generative AI: Concepts and Applications

What is Generative AI?

Generative AI refers to a class of artificial intelligence systems that can create new, original content — such as text, images, audio, video, or code — rather than simply analysing or classifying existing data. These models learn the underlying patterns and structure of their training data and then use that knowledge to generate novel outputs that resemble what they were trained on.

Key Concepts

- **Generative Models:** Learn the probability distribution of training data (e.g., GANs, VAEs, Diffusion Models, Transformer-based models).
- **Training Data:** Large datasets (text corpora, image sets, audio libraries) used to teach the model patterns.
- **Prompt:** The input/instruction given by a user that guides what the model generates.
- **Fine-tuning:** Adapting a pre-trained generative model to a specific task or domain.
- **Tokens:** Basic units (words/sub-words) that text-based generative models process and generate.

Core Techniques Behind Generative AI

- **Generative Adversarial Networks (GANs)** — a generator and discriminator compete to produce realistic outputs.
- **Variational Autoencoders (VAEs)** — encode data into a latent space and decode it back to generate variations.
- **Diffusion Models** — generate data by gradually removing noise from random input (used in image generation).
- **Transformer Architecture** — the backbone of modern large language models like GPT.

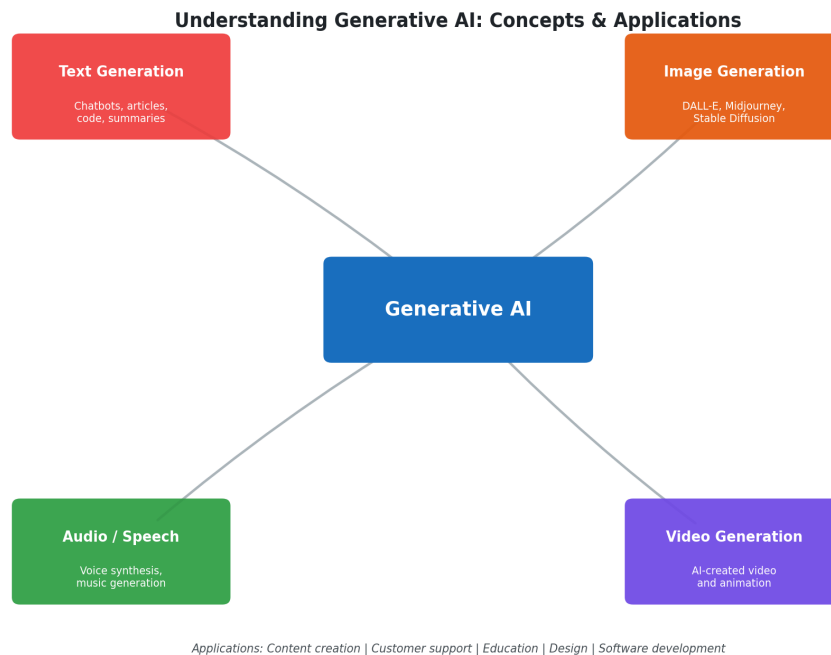


Figure 1: Generative AI — core categories and applications

Applications of Generative AI

- Content Creation: Articles, marketing copy, poetry, scripts.
- Software Development: Code generation, debugging assistance, documentation.
- Design & Art: AI-generated images, logos, and illustrations.
- Education: Personalised tutoring, question generation, summarisation.
- Healthcare: Drug discovery, synthetic medical data, report drafting.
- Customer Support: Chatbots and virtual assistants.

2. Evolution of Language Models: From Rule-Based Systems to LLMs

Language models have evolved dramatically over the decades — from simple hand-coded rules to today's massive neural networks capable of human-like conversation.

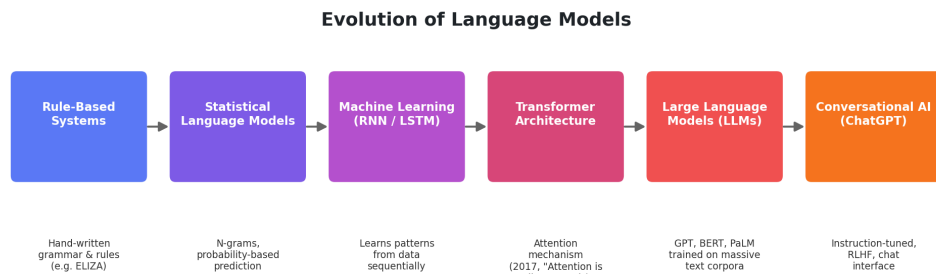


Figure 2: Timeline — evolution of language models

Stage-by-Stage Summary

- Rule-Based Systems (1950s–80s): Relied on hand-written grammar rules and pattern matching. Example: ELIZA (1966), an early chatbot that mimicked a psychotherapist using scripted responses.
- Statistical Language Models (1980s–00s): Used probability theory (n-grams) to predict the next word based on frequency of word sequences in a corpus. Limited by inability to capture long-range context.
- Machine Learning Era — RNNs/LSTMs (2000s–2015): Neural networks that could process sequences and retain some memory of previous words, improving context handling but struggling with very long sequences.
- Transformer Architecture (2017): Introduced in the paper “Attention Is All You Need.” Uses a self-attention mechanism to weigh the importance of all words in a sequence simultaneously, enabling parallel processing and better long-range context understanding.
- Large Language Models – LLMs (2018–present): Models like BERT, GPT-2/3/4, and PaLM are trained on massive internet-scale text corpora with billions of parameters, enabling broad language understanding and generation.
- Conversational AI – ChatGPT (2022–present): LLMs fine-tuned specifically for dialogue using instruction tuning and Reinforcement Learning from Human Feedback (RLHF), making them safer and more helpful in conversation.

Why This Evolution Matters

- Each stage removed a major limitation of the previous one (manual rules → data-driven → contextual understanding → scale).
- The shift to Transformers enabled today's foundation models trained once and adapted to many tasks.

3. Introduction to ChatGPT: Features, Capabilities, and Use Cases

What is ChatGPT?

ChatGPT is a conversational AI system developed by OpenAI, built on the GPT (Generative Pre-trained Transformer) family of large language models. It is fine-tuned using supervised learning and Reinforcement Learning from Human Feedback (RLHF) to hold natural, helpful, multi-turn conversations.

Key Features

- Natural Language Understanding: Interprets user intent across varied phrasing and context.
- Multi-turn Conversation: Remembers earlier parts of the conversation to maintain context.
- Text Generation: Produces essays, summaries, emails, stories, and code.
- Language Translation: Converts text between multiple languages.
- Reasoning & Problem-Solving: Handles logical, mathematical, and analytical tasks step-by-step.
- Customisation: Can be guided using system prompts, custom instructions, and fine-tuning.

Capabilities

- Summarising long documents into concise notes.
- Drafting and debugging code in multiple programming languages.

- Answering questions across a wide range of domains.
- Brainstorming ideas and generating creative content.
- Simulating roleplay and interactive dialogue scenarios.

Common Use Cases

- Education: Tutoring, explaining concepts, generating practice questions.
- Business: Drafting reports, emails, presentations, and meeting summaries.
- Software Development: Code generation, review, and documentation.
- Customer Service: Automated, context-aware support chatbots.
- Research: Literature review assistance and idea exploration.

Limitations to Be Aware Of

- Can produce plausible-sounding but incorrect information (“hallucination”).
- Knowledge is limited to its training data cut-off unless connected to external tools.
- Lacks true understanding — generates responses based on learned patterns, not reasoning from first principles.

4. Navigating Ethical Considerations: AI-Generated Content and Responsible Usage

Why Ethics Matter in Generative AI

As generative AI tools become widely accessible, their ability to create realistic and persuasive content raises important ethical, legal, and social concerns that users and developers must navigate responsibly.

Key Ethical Concerns

- Misinformation & Deepfakes: AI-generated text, images, or videos can be used to spread false information or impersonate real people.
- Bias & Fairness: Models can inherit and amplify biases present in their training data, leading to unfair or discriminatory outputs.
- Copyright & Intellectual Property: Questions around ownership of AI-generated content and use of copyrighted material in training data.
- Privacy: Risk of models memorising and revealing sensitive personal information from training data.
- Academic & Professional Integrity: Concerns over plagiarism and over-reliance on AI for original work.
- Job Displacement: Automation of content-creation tasks may impact certain roles and industries.
- Accountability: Determining responsibility when AI-generated content causes harm.

Principles for Responsible Usage

- Transparency: Disclose when content is AI-generated, especially in journalism, academia, and marketing.
- Human Oversight: Keep a human in the loop to review and validate AI outputs before use.
- Fact-Checking: Verify AI-generated claims against reliable sources.

- Bias Mitigation: Regularly audit models and outputs for fairness across demographics.
- Respecting IP Rights: Avoid using AI to reproduce copyrighted material without permission.
- Data Privacy: Avoid inputting sensitive personal or confidential data into public AI tools.
- Following Guidelines: Adhere to organisational and regulatory AI usage policies.

Emerging Regulatory Landscape

- Governments and organisations worldwide are developing AI governance frameworks (e.g., EU AI Act) to ensure safe and ethical deployment.
- Industry bodies are promoting AI usage guidelines around content labelling and accountability.